

Quasi-amorphous crystals

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Quasi-brittle yielding is a salient feature of well-annealed glassy materials. Here we show that the same behavior is characteristic of plastic yielding in perfect crystals after they experience mechanically driven elastic instability leading to massive nucleation of dislocations. We argue that such ‘preparation’ effectively converts the atomic configuration from crystalline to quasi-amorphous. To understand the nature of subsequent quasi-brittle yield we study mechanical response of a prototypical 2D crystal subjected to quasistatic loading. We show that both pre- and post-yield dislocation avalanches exhibit power law statistics with matching exponents whose value is indicative of marginal stability.

The complexity of plastic yield in crystalline solids is exacerbated by the fact that both pre- and post-yield plastic flows often exhibit intermittent fluctuations and scale-free organization of dislocations [1–10]. These features are also typical for amorphous plasticity [11–20]. To build a conceptual link between the two phenomena, we study in this paper a relatively transparent prototypical example of a 2D square crystal subjected to quasistatic loading.

To prepare a ‘generic’ crystal we start with a pristine dislocation-free configuration and bring it to the point of mechanical instability which is resolved through massive dislocation nucleation. We argue that such ‘preparation’ effectively converts the atomic configuration from crystalline to quasi-amorphous [21–23]. We show that the subsequent plastic flow begins with a hardening stage, which involves a superposition of spatially localized dislocation rearrangements. Such ‘microplasticity’ regime is terminated abruptly with a quasi-brittle system-size event leading to the formation of multi-grain pattern arranged in the form of global shear bands. The post-yield plastic flow is characterized by a stress plateau with superimposed fluctuations representing broadly distributed intermittent dislocation avalanches. Similar quasi-brittle yield is known to be characteristic of plastic response of well-annealed glasses [24–32]. While the implied correspondence may look surprising in view of the absence of dislocations in amorphous plasticity, we recall that behind both phenomena is an evolving elastic incompatibility [33–37] with underlying elementary mechanical events represented by shear eigenstrains [38–40].

Our conclusions are based on a detailed study of dislocation dynamics in a prototypical square crystal subjected to quasistatic loading by shear strain applied along one of the slip directions. Our main tool is a novel mesoscopic tensorial model (MTM) of crystal plasticity representing a conceptual trade-off between continuum and atomic descriptions [41–48]. Crucially, it allows one to capture the intermittent nature of dislocation avalanches [49, 50]. In contrast, other computational approaches [51–76] struggle to simultaneously reproduce the correct statistics of plastic fluctuations, account for large lattice rotations, and adequately represent the crystallographic nature of lattice-invariant shears [77–81].

The MTM approach assumes that mesoscale material elements experience an effective periodic energy landscape ac-

counting for lattice-invariant shears. Within this Landau-type continuum framework with infinitely many equivalent energy wells [41–48, 82–91], plastically deformed crystals emerge as multi-phase mixtures of equivalent phases. Despite operating with engineering concepts of stress and strain, MTM correctly captures both long-range elastic interactions between dislocations and their short-range interactions, including topological changes such as nucleation and annihilation [92–101].

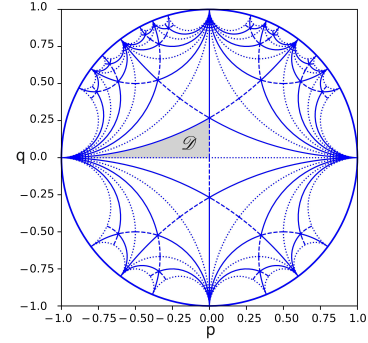


FIG. 1. $GL(2, \mathbb{Z})$ -induced tessellation of the configuration space of metric tensors with $\det \mathbf{C} = 1$ stereographically projected on a Poincaré disk; see [102] for details. The domain $\mathcal{D}$ highlighted in gray is a minimal periodicity domain. Black squares mark equivalent representations of the same square lattice.

Suppose that $\mathbf{y} = \mathbf{y}(\mathbf{x})$ is the deformation where $\mathbf{y}$ and $\mathbf{x}$ are positions of material point in the current and reference configurations. Due to frame indifference constraint, the strain energy density of an elastic solid can depend on the deformation gradient $\mathbf{F} = \nabla \mathbf{y}$, only through the metric tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ which then plays the role of the order parameter.

The elastic energy density $\varphi = \varphi(\mathbf{C})$ should also respect the symmetry of an underlying Bravais lattices going beyond the conventional continuum mechanics point group. Thus, in the corresponding 2D continuum theory one should require that $\varphi(\mathbf{C}) = \varphi(\mathbf{m}^T \mathbf{C} \mathbf{m})$, where $\mathbf{m}$ belongs to $GL(2, \mathbb{Z}) = \{\mathbf{m}, m_{IJ} \in \mathbb{Z}, \det(\mathbf{m}) = \pm 1\}$. In view of this global symmetry, the surface $\det \mathbf{C} = 1$ in the 3D space (C_{11}, C_{22}, C_{12}) is tessellated into an infinite number of equivalent periodicity domains. To visualize this tessellation, it is convenient to project stereographically the infinite surface $\det \mathbf{C} = 1$ on a unit (Poincaré) disk, see Fig. 1 where the equivalent square

lattices are indicated by small black squares. **WE REMOVED THE SQUARE DOTS** The structure of the minimal tensorial periodicity domain $\mathcal{D}$, is discussed in [102]; if the energy density is known in one of such domains, it can be extended to the whole configurational space by symmetry.

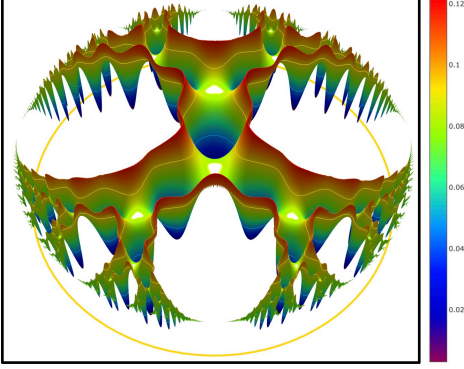


FIG. 2. Energy landscape in the space of metric tensors (Poincaré) disk shown in Fig. 1). Color indicates the energy level with only the lowest energy levels shown. The deepest energy wells correspond to equivalent representations of the same square lattice.

In our numerical experiments we used the simplest function $\varphi(\mathbf{C})$ with required periodicity discussed in [102]. In Fig. 2 we illustrate the corresponding multi-well energy landscape with bottoms of the wells corresponding to lattice invariant deformations. One can show that such a landscape contains a network of low energy valleys along the directions representing plastic ‘mechanisms’, simple shears parallel to crystallographic slip planes. Along these valleys the energy wells correspond to the equivalent replicas of the reference square lattice.

The presence of a complex system of energy barriers separating individual quantized lattice invariant shears makes the function $\varphi(\mathbf{C})$ highly nonconvex. Therefore, as in every Landau type continuum theory, the corresponding scale free problem is highly degenerate. Instead of conventional Ginzburg-type gradient regularization, we introduce in MTM a uniform mesoscale grid and attribute elastic response to discrete elements whose linear size h is then viewed as a physical parameter of the model. In other words, we reduce the space of admissible deformations to a finite dimensional set of compatible, piece-wise affine mappings, see [102] for details.

It is clear that the ensuing mechanically driven system will experience a rich repertoire of instabilities. In particular, under quasistatic loading conditions the incremental elastic energy minimization will lead to a succession of jump discontinuities representing elastic branch switching events. The corresponding energy losses will then mimic dissipative processes which will merge in the continuum limit producing what is known as rate-independent plasticity [103].

In our numerical experiments, a square crystal sample was discretized with $N \times N$ nodes, which were triangulated into $2N^2$ finite elements. We used $N = 400$. The details of the incremental energy minimization algorithm, the loading

protocol and the boundary conditions are discussed in [102]. We loaded our athermal system quasi-statically by applying the affine displacement field $\mathbf{u}(\alpha, \mathbf{x}) = (\bar{\mathbf{F}}(\alpha) - \mathbb{1})\mathbf{x}$, with $\bar{\mathbf{F}}(\alpha) = \mathbf{I} + \alpha \mathbf{e}_1 \otimes \mathbf{e}_2$, where $\mathbf{e}_i$ is the orthonormal basis of the reference lattice. Such loading was chosen to represent simple shear along one of the principal slip directions, with shear amplitude α serving as the loading parameter. By changing this parameter in strain increments of order 10^{-6} , we advanced the loading, and after each increment the displacement field was updated by the energy minimization algorithm.

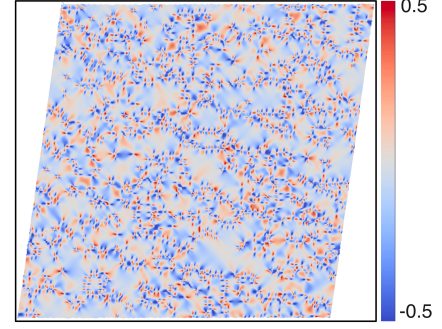


FIG. 3. Dislocated configuration emerging after a homogeneous lattice is mechanically driven to the threshold of elastic instability. Colors indicate the level of the shear component of the Cauchy stress tensor.

To ‘prepare’ a generic sample we first loaded a pristine (defect-free) crystal. The associated purely elastic homogeneous deformation became unstable at a critical value of the loading parameter $\alpha = \alpha_c \approx 0.138$, which is well approximated by the value corresponding to the loss of positive definiteness of the acoustic tensor, see [102, 104–106]. The breakdown of an elastic state took the form of a major system-size inelastic deformation transforming a perfect crystal into a dislocation-rich quasi-amorphous configuration, see Fig. 3 where the highest stress levels correspond to the location of dislocation cores. Note that the stabilization of the global dynamic restructuring process can be expected to produce only a marginally stable state.

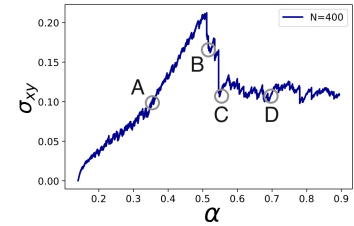


FIG. 4. Stress-strain response of a quasi-amorphous crystal shown in Fig. 3.

Having ‘prepared’ a generic sample, we continued to load it using the same athermal quasistatic loading protocol. The ensuing stress-strain response of our quasi-amorphous crystal is illustrated in Fig. 4. Note the presence of irregularly placed elastic branches interrupted by abrupt stress drops; the latter represent intermittent plastic avalanches which involve partial

or complete unlocking of dislocation structures. One of these avalanche-type events, representing a major multi-stage stress drop, stands out, which suggests that there are two different regimes. While the first regime is characterized by plastic hardening, the second regime, represented by a stress plateau, suggests quasi-stationary plastic flow, see Fig. 4. It is natural to interpret the abrupt system-size event dividing these two regimes as a quasi-brittle plastic yield.

Fig. 5 shows the stress field in a typical pre-yield state. Its comparison to Fig. 3 suggests that the restructuring is relatively minor with mostly preexisting mobile dislocations moving between the locking sites formed by immobilized dislocations. In other words, dislocations appear to be confined inside the effective cages where they can pile up but from where they can only rarely escape by breaking the existing locks. Overall, such a quasi-elastic regime preceding macroscopic quasi-brittle yield can be characterized as 'microplasticity' which is known to be characteristic of pre-yield behavior for both crystalline and amorphous solids [4, 107–111].

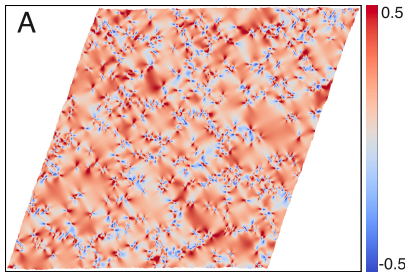


FIG. 5. Dislocated configuration in a typical pre-yield state characterizing the stage of 'microplasticity'. The state *A* is marked in Fig. 4. Colors indicate the level of the shear component of the Cauchy stress tensor.

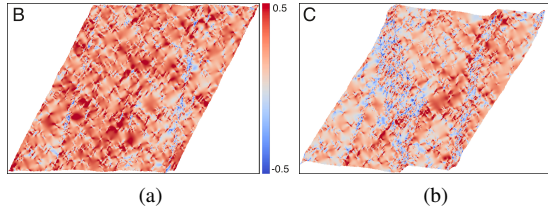


FIG. 6. Dislocated configuration characterizing different stages of unfolding of the quasi-brittle system size event. The states *B* and *C* are marked in Fig. 4. Colors indicate the level of the shear component of the Cauchy stress tensor.

The quasi-elastic regime ends with discontinuous yield which is highly reminiscent of mechanical response of well annealed 'brittle' glasses; the phenomenon of plastic brittleness is also a characteristic feature of sub-micron crystal samples with high initial purity [112–122]. As we see in Fig. 4, the global restructuring event takes the form of a two-stage transition $A \rightarrow B \rightarrow C$. The snapshots of the spatial dislocation microconfigurations in the states *B* and *C* are shown in Fig. 6 where we already see the evidence of a collective dislocation activity which leads to the emergence of global structures in

the form of system-spanning shear bands.

Additional important details can be seen in Fig. 7 where we illustrate the strain-energy density field in state *C*. Here, for instance, one can distinguish dislocations of different sign via Delaunay triangulation (see inset in Fig. 7): nodes with five (seven) neighbors are shown in blue (red), and dislocations correspond to adjacent pairs where the ordering 5-7 or 7-5 determines the sign. It also becomes visible that the shift from 'local' to 'global' dislocation rearrangements leads to the development of dislocation-free rotated patches of the original lattice forming the polycrystalline grain texture inside the shear bands. The misorientation of neighboring grains is controlled by the overall compatibility of the deformation field. The emergence of special highly optimized grain boundaries, in particular, the ubiquitous presence in such patterns of low energy $\Sigma 5$ grain boundaries, has been discussed in [47], see also [123–125].

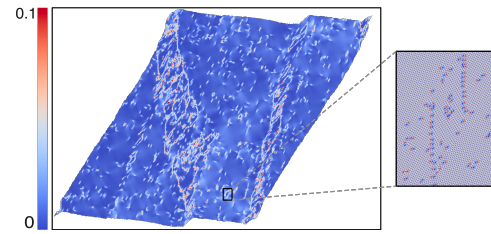


FIG. 7. The micro-configuration configuration corresponding to the state *C* marked in Fig. 4. Colors indicate the magnitude of strain-energy density. The inset shows a single grain and details the dislocation structure of the grain boundaries.

The post-yield regime characterized by relative stabilization of the average stress level can be interpreted as a quasi-stationary non-equilibrium steady state. A representative snapshot of the microconfiguration of the crystal in state *D* is shown in Fig. 8 where we see further development of the polycrystalline grain texture. The intermittent avalanches characterizing this regime represent bigger grains occasionally merging to form larger grains while small grains continue to nucleate. In this way the process of self-organization of dislocations within the widening shear bands continues to produce additional scales developing into a global hierarchical structure. Overall the system transitions from isolated dislocation motion to collective dislocation behavior as it gains access to a much broader repertoire of large-scale relaxation mechanisms. A detailed analysis of the avalanche structure shows that while in the pre-yield regime plastification occurs primarily in isolated, linear arrangements of transformed elements showing minimal branching, the configuration of a typical post-yield avalanche reveals extended plastified regions with complex branching.

The value of the power-law exponent $\epsilon = 1.0$ is a signature of an archetypically 'wild' crystal plasticity in the sense of [50]. The same value was obtained in discrete dislocation dynamics studies of disorder-free 2D samples containing a fixed number of preexisting dislocations [63, 126] and also recorded in numerical experiments on pristine crystals using a scalar version of the MTM [48]. The exponent

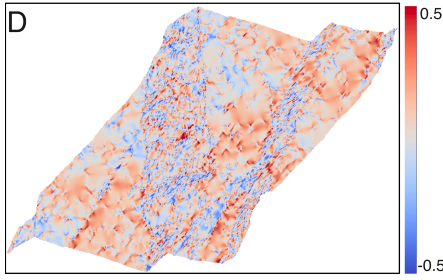


FIG. 8. Mature shear bands in the regime of stationary post-yield plastic flow represented by the state D marked in Fig. 4. Colors indicate the level of the shear component of the Cauchy stress tensor.

$\epsilon = 1.0$ has been previously interpreted in crystal plasticity framework as representing either dislocation jamming or self-induced glassiness [21, 127–129]. The latter conclusion is corroborated by the fact that the same value of the exponent is routinely recorded in the studies of plastic flows in structural glasses [13, 130, 131].

A theoretical understanding of the relation between the emergence of the power law exponent value $\epsilon = 1.0$ and the structure of the underlying energy landscapes has been developed in the theory of spin glasses, where it was shown that the hierarchical (ultrametric) organization of energy wells in the phase space results in an intermittent, scale-free response to quasistatic deformation [132–135] and that in the mean-field limit the associated distribution of energy avalanches must be of a power law with $\epsilon = 1.0$. In this context it was also understood that the reason behind this value of the exponent is the marginal stability of the system [132, 136, 137].

Since glasses represent liquids at a quenching threshold, their marginal stability is not surprising, for instance, similar marginality is exhibited by jammed states of granular matter. If such marginally rigid systems are mechanically driven, the proximity to unstable modes makes the mechanical response inherently *intermittent, effectively* mixing statics (stability) and dynamics (instability). Recall that our ‘quasi-amorphous crystal’ also emerged from arrested dynamics with the system being stabilized close to the threshold of mechanical stability. One can argue that in such cases the self-generated disorder brings the system directly from solid to glassy state. Since we observe the same value of exponent $\epsilon = 1.0$ in both pre- and post-yield regimes, we may conclude that the achieved marginality is not even affected by the quasi-brittle yielding event. This may mean that the brittle yielding does not compromise the global organization of the emerged energy landscape while obviously affecting the reachability of its different subdomains [138–140].

Note next that the pre-yielding value of the cutoff parameter $\lambda = 0.0230$ is drastically different from its post-yielding value $\lambda = 0.0076$. The approximately three-fold decrease of the value of λ after yielding indicates that large energy drops become substantially more probable in the post-yield regime. The emergent collective behavior in this regime manifests itself in a much broader extent of avalanche activity reaching the size of the system and in this sense the system shows the

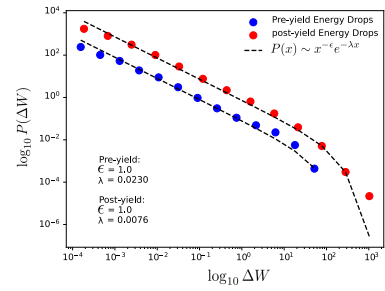


FIG. 9. Probability distribution of energy drops during avalanches in pre- and post-yield regimes.

signs of divergence of the characteristic length $l \sim 1/\lambda$.

This conjecture is supported by the observation that shear bands in the post-yield regime traverse the whole computational domain which is in stark contrast to unequivocally subextensive localized dislocation rearrangements taking place in the pre-yield regime. The physical nature of the implied criticality is presently actively debated [48, 141–145].

Finally we mention that if intermittent fluctuations in amorphous plasticity are usually modeled in terms of elasto-plastic models [24, 131, 146, 147], the MTM approach to crystal plasticity is very similar, basically operating within the same finite element setting. The difference is that the phenomenological yield thresholds are modeled in MTM by elastic instabilities reflecting the structure of the geometrically conditioned energy landscape while the fixed elastic propagators are replaced by the solution ‘on the fly’ of the corresponding nonlinear elasticity problems. The implied nonlinearity is also of universal nature as it originates from the geometrically exact description of deformation allowing one to handle adequately both finite rotations and elastic compatibility.

In this paper we performed a series of numerical experiments aimed at deeper understanding of the phenomenon of plastic yield in model 2D crystals in the limit of zero strain rate and negligible thermal effects. While it is known that quasi-brittle yielding is a salient feature of well-annealed glassy materials, we showed that generic crystals can exhibit remarkably similar phenomenology after the necessary level of annealed disorder emerges from dislocation self-organization after the breakdown of an affine elastic configuration. One of our main findings is that in the ensuing quasi-amorphous crystals both pre- and post-yield avalanches exhibit power law statistics with matching exponents indicative of hierarchical structure of the underlying energy landscape and marginal stability of the associated mechanical system. *It would also be intriguing to explore whether crystals with higher triangular symmetry exhibit similar behavior and to assess whether the specific choice of strain-energy density functional plays a crucial role in these phenomena.* More work is needed to turn our initial insights into an understanding of similar phenomena in realistic 3D crystals with HCP, FCC and BCC symmetry. In particular, only a 3D model will distinguish adequately between edge and screw dislocations, and will be suitable to handle realistic dislocation entanglements. It will then be able to capture such important physical effects

as dislocation climb, cross slip and forest hardening.

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